

Functional specifications

Description Eartags module
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Changes of document

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General description

In eartags module activities regarding new and duplicate eartags are supported. Veterinary department (VD) uses eartags module to keep track of issued eartags, majority of other users uses it to place eartags orders and to track those orders until delivery. This is also basic and most used module for eartags suppliers (at the moment VD is the only eartags supplier in the country, however module allows more suppliers).

Processes

Issuing of new eartags – when needed, VD can issue new ear tag numbers, which will be later printed on eartags. The procedure is as follows:

1. on web form VD user selects “generate new ear tag numbers”
2. a new web form is opened, where he/she has to enter the following data:
 - a. number of ear tag numbers to be generated
 - b. description (usually a number of VD’s internal document) – this data is optional
3. when “proceed” button is pressed (and confirmed via conformation box), specified amount of ear tag numbers are generated and placed into appropriate tables with the status “NEW” (see data/business rules section for detail about generating ear tag numbers)
4. when step 3 is finished, user gets new web page where statistic data is shown. In case of an error, appropriate error message is shown and all changes are rolled back

Defining supplier contingent of new eartags – each supplier (at the beginning the only supplier will be VD) can get contingent(s) of ear tag numbers, from which he/she can print eartags. Those ear tag numbers are sent from VD to supplier in form of a flat (txt) file which has the following structure:

- name: et_cont_yyyymmdd_x.txt
- inside text file, each row has two (tab delimited) texts: *state* (MK) and *ear tag number* (8 digits)

Procedure for generating supplier contingent is as follows:

1. on appropriate web form VD user selects “generate supplier ear tags contingent” option a new web form is opened, where he/she has to enter/select the following data:
 - a. select the supplier
 - b. enter number of ear tag numbers to be assigned to supplier
 - c. description (usually a number of VD’s internal document) – entering this data is optional
2. when “proceed” button is pressed (and confirmed via conformation box), specified number of ear tag numbers are into appropriate tables receive status “AVAILABLE”. They are also marked that they “belong” to specified supplier (see data/business rules section for detail about generating ear tag numbers)
3. when step 3 is finished, user is asked to save txt file. In case of an error, appropriate error message is shown and all changes are rolled back
4. for each ear tag supplier contingent can be set only once, however output file (which was received in step 4) is stored into database and can be downloaded (via appropriate web form) again if needed.

Placing order for new eartags – from “ordinary” user’s point of view, this is one of core functionality of eartags module. Procedure is as follows:

1. User selects “placing orders for new eartags” option from main web page
2. a new web form opens where user must/can enter the following data:
 - a. select a supplier of eartags – if there is only one supplier than this step is optional (user doesn’t have option of selecting supplier – this part of web page is not shown to him/her). In this case system already knows which supplier to select
 - b. enter number of eartags to be placed on order (this data is mandatory)
 - c. enter farm id – entering this data is optional and is only available to suppliers, veterinary stations and VD. It is used for signaling who is payer of order and where should eartags be sent. In case that user of the system comes from a farm, than he/she can not enter this data (this part of web page is hidden), because he/she is an end user/payer of eartags (for detailed explanation see business rules section)
3. when “proceed” button is pressed, a list of business rules is checked (see business rules section) and then a new web page is shown. If entered data passed all business rules then order preview is shown, based on data entered on previous web page and system data (for example organization of user who placed an order). In order to proceed, user has to check two checkboxes (statement that he/she agrees that data on order preview are correct and that person/organization which is listed as receiver of eartags can be charged for those eartags) and press button “proceed”
4. order is placed (data written to appropriate tables) and user has option to print order or place another order (back to step 2)

Placing order for duplicate eartags – on the field eartags can be lost/destroyed/.... In such cases duplicate eartags are needed. Procedure is very similar to the one of placing orders for new eartags, however there are different business rules and ear tag numbers are known at the time of placing order. Detailed procedure is as follows:

1. User selects “placing orders for duplicate eartags” option from main web page
2. a new web form opens where user must/can enter the following data:
 - a. select a supplier of eartags – if there is only one supplier than this step is optional (user doesn’t have option of selecting supplier – this part of web page is not shown to him/her). In this case system already knows which supplier to select
 - b. specify state, ear tag number (id animal) and type of duplicate (single or for both ears) - this data is mandatory; optionally user can enter additional description for each ear tag
 - c. enter farm id – entering this data is optional and is only available to suppliers, veterinary stations and VD. It is used for signaling who is payer of order and where should eartags be sent. In case that user of the system comes from a farm, than he/she can not enter this data (this part of web page is hidden), because he/she is an end user/payer of eartags (for detailed explanation see business rules section)
 - d. enter address where eartags shall be sent to - entering this data is also optional and is only available to suppliers, veterinary stations and VD. It is used for signaling where eartags should be sent. If user of the system comes from a farm, than he/she can not enter this data (this part of web page is hidden), because he/she is an end user/payer of eartags (for detailed explanation see business rules section). If user entered farm id, then this data (address) is ignored
 - e. user has also option to append in step b. entered eartags to existing order (see business rules for detail)
3. when “proceed” button is pressed, a list of business rules is checked (see business rules section) and then a new web page is shown. If entered data passed all business rules then order preview is shown, based on data entered on previous web page and system data (for example organization of user who placed an order). In order to proceed, user has to check two checkboxes (statement that he/she agrees that data on order preview are correct and that person/organization which is listed as receiver of eartags can be charged for those eartags) and press button “proceed”
4. order is placed (data written to appropriate tables) and user has option to print order or place another order (back to step 2)

Supplier’s retrieval of placed orders (new eartags) – supplier has to collect orders for new eartags which were placed for him and print eartags according to the following procedure:

1. select “collect orders of new eartags” option
2. appropriate file is generated. For each order the following data is included (all orders are placed into one file):
 - a. order’s info (where to send eartags, who will pay)
 - b. list of ear tag numbers on order (see business rules section for detail about which (new) ear tag numbers are automatically put on order)
3. Collected orders and corresponding eartags get status “COLLECTED” – signaling that supplier has required data and that eartags are in process of printing and delivery. If any error occur appropriate error message is shown to user and all changes are rolled back.

Supplier’s retrieval of placed orders (duplicate eartags) – supplier has to collect orders for duplicate eartags which were placed for him and print eartags according to the following procedure:

1. select “collect orders of duplicate eartags” option
2. appropriate file is generated. For each order the following data is included (all orders are placed into one file):
 - a. order’s info (where to send eartags, who will pay)
 - b. list of ear tag numbers on order – eartags are placed on order at the moment when user puts order into system
3. Collected orders and corresponding eartags get status “COLLECTED” – signaling that supplier has required data and that eartags are in process of printing and delivery. If any error occurs appropriate error message is shown to user and all changes are rolled back.

Reviewing/managing placed orders – placed orders can be viewed, for each order details can be seen. Users can also cancel their orders (see business rules for details about canceling orders). Within this sub module, suppliers are also able to confirm delivery of eartags. From supplier’s point of view, the following process of viewing/managing orders for eartags is in place:

1. select “view/manage eartags orders”
2. a new parameter web form is shown where user can limit search according to his/hers needs
3. after parameters were entered and “proceed” button was pressed on parameter form, a list of corresponding (according to given parameters) orders is shown. Implicit limitations in regard of which orders can be seen are applied, based on privileges user has (see business rules for detail). On this list supplier can observe general information regarding each order that was placed for her/him. From this list the following options are available:
 - a. Confirmation of delivery of eartags – if order has been collected, then a checkbox next to order number can be checked. At the end of the page there is also button “Select all”, which sets checkbox for all

supplier's collected orders. Checking of checkbox next to order number means that this order has been delivered. Further actions/steps are:

- i. When button proceed is pressed a new web page opens where list of delivered orders (those that were checked on previous screen) is shown. All orders on this list and corresponding eartags automatically get status "DELIVERED".
- ii. If any error occurs, all changes are rolled back and appropriate error message is displayed.
- b. View details regarding specific order – when user clicks on order number a new web page is opened. On this form the following additional actions are available:
 - i. User can cancel order (if conditions are met - see business rules section for details) – in this case he/she must press button "CANCEL ORDER". A new web page will be opened where data regarding cancelled order is shown. Order gets status "CANCELLED". If any error occur then appropriate error message is shown and all changes are rolled back.
 - ii. User can cancel specific ear tag number (if conditions are met - see business rules section for details) – in this case he/she must press button "CANCEL EARTAG". A new web page will be opened where data regarding cancelled order is shown. Ear gets status "CANCELLED". If there are any errors then appropriate error message is shown and all changes are rolled back.
 - iii. For each ear tag number which is on list, it is possible (if number of ear tag is clicked) to see history of previous orders for this ear tag.

Note: on orders for new eartags no eartags are on order until supplier collects order.

Data validation and business rules

Generating of ear tag numbers

- ear tag number has 8 digits where the last digit is a check digit
- check digit is automatically generated by this formula:
$$\text{Check digit} = \text{mod } 10 \text{ of } \text{sum of } 3*\text{dig1} + 5*\text{dig2} + 7*\text{dig3} + 11*\text{dig4} + 13*\text{dig5} + 17*\text{dig6} + 19*\text{dig7}$$
- ear tag numbers start from 10000001 – in that way a fixed length of ear tag number is assured

Rules when ordering new eartags:

- user can insert data (not just view)
- if farm id is entered then user has to have privilege to enter data for that farm
- if farm id is specified, then number of new eartags that can be ordered is computed according to the formula:
$$\text{no_eartags_to be ordered} = \text{number of female animals on farm} - \text{no. of remaining eartags for this farm from previous orders}$$
- if id farm is entered, then this farm must be valid for ordering (for example for factious farm id, slaughter house etc. eartags can not be ordered)
- when farm id is entered, the following tim-related conditions have to be met:
 - the has to be a 120 (this is system parameter and can be changed if needed) day gap between current order and last (valid- not cancelled) order for this farm id
 - for a farm maximum of 4 orders of new eartags per year can be placed
- the same order can be entered only once (to prevent errors based on back/reload buttons on web browsers)

Rules when ordering duplicate eartags:

- the same order can be entered only once (to prevent errors based on back/reload buttons on web browsers)
- animal for which request for duplicate eartag is issued has to be alive
- if farm id is entered then for animal for which request for duplicate eartag is issued has to be on that farm
- if farm id is entered then user has to have privilege to enter data for that farm
- if id farm is entered, then this farm must be valid for ordering (for example for factious farm id, slaughter house etc. eartags can not be ordered)

Putting new ear tag numbers on order automatically – when supplier takes over (collects) orders that were placed for him/her, system automatically “puts” ear tag numbers on appropriate orders according to the following rule:

- for each order in collection check how many eartags were ordered
- for the number of ordered eartags, go through the list of still available ear tag numbers, which were assigned to supplier
- put those ear tag numbers on order (and in output file). Sort ear tag numbers on order ascending
- if anything fails (supplier doesn’t have enough ear tag numbers available, or some system error occurs) – print out appropriate error message and roll back all changes that have been made

Append additional (duplicate) eartags to existing order – if user wants to do this operation, the following conditions have to be met:

- “original” order is his/hers
- “original” order is still valid (not cancelled) and supplier hasn’t collected this order for printing eartags yet
- all other business rules regarding ordering duplicate eartags have passed successfully

Limitations in regard of viewing orders for eartags (privileges):

- “ordinary” user can see at most those orders for eartags, that were placed by him/her or hers/his organization
- supplier can see only those orders that were placed for him/her (as a supplier of eartags) and not for other suppliers
- VD can see all orders without (privilege based) restrictions.

Rules for canceling of order for eartags:

- Order can be cancelled only by user who placed an order for eartags or VD
- Meanwhile no other activities for this order took place (for example supplier hasn’t collected this order yet)
- The same rules apply for a specific ear tag on this order (it is possible to cancel ordering of specific ear tag without canceling whole order)

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- Specific eartag can be cancelled also in period when delivery has been made, but before first tagging with this eartag. Usually data related to this kind of cancellation will be received from PDA. This happens when eartag is destroyed at first tagging (but before registration is entered)

